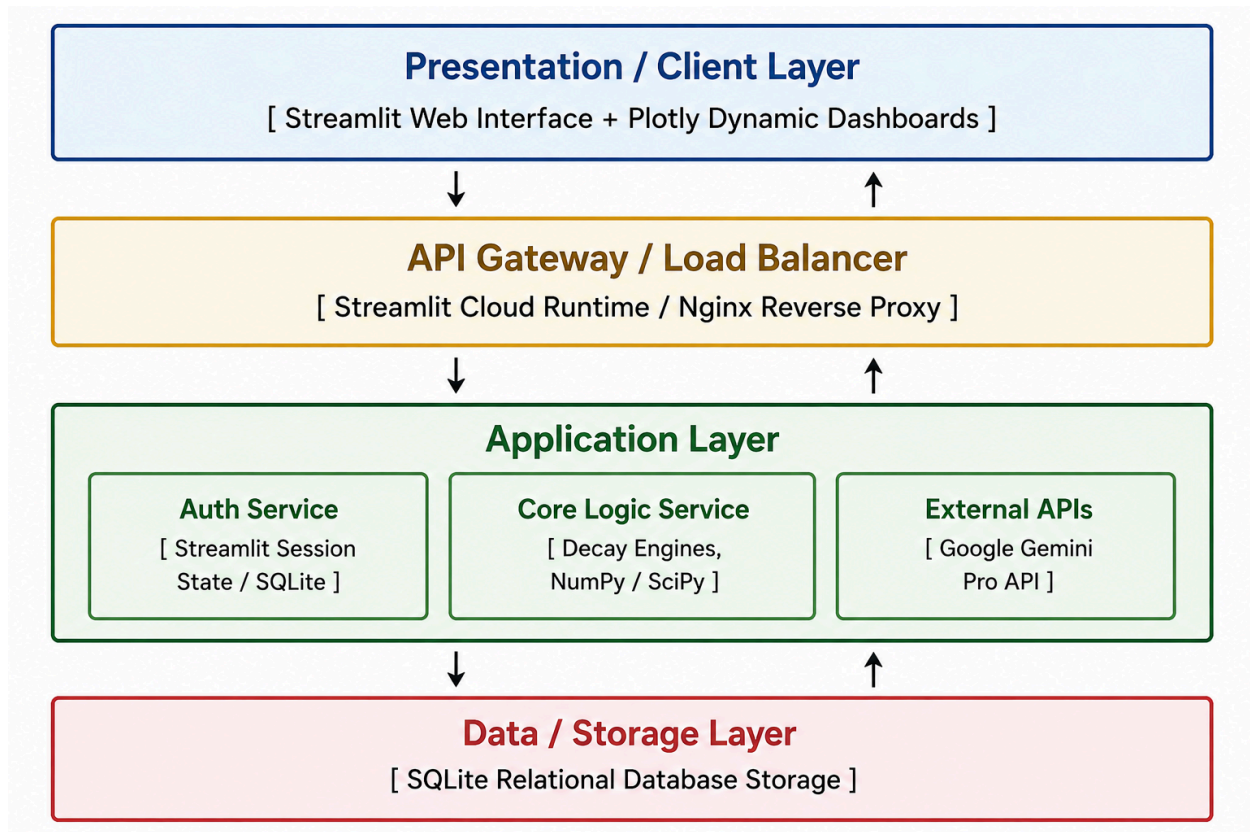


Solution Architecture

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architecture diagram.



Component Description Table

Component Name	Description/Role in Architecture	Technologies Used
Presentation Layer	Renders the interactive frontend interface, taking user input, displaying mathematical forgetting curves, and surfacing adaptive quizzes.	Streamlit UI, Plotly
API Gateway	Routes traffic between the browser and backend services, securing endpoints and maintaining session communication.	Streamlit Cloud / Nginx Reverse Proxy
Microservices / Application Services	Executes core calculations (Ebbinghaus decay functions), handles quiz generation requests, and manages session state.	Python 3.10+, NumPy, SciPy, Google Gemini API
Database	Stores persistent records including user profiles, quiz performance history, and memory retention state data.	SQLite